

## Example 1: Sweet Apple Acres

Applejack has come to us: “Between the Red Delicious row, the Honeycrisp row, and that new Zap Apple row Granny Smith insisted on plantin’, I reckon one of ‘em oughta be doin’ better than the others. Can you tell me if they’re really any different, or am I just imaginin’ things?”

- Glimpse of data:

```
# A tibble: 3 × 2
  row      bushels
<chr>    <dbl>
1 Red Delicious Row  42.2
2 Red Delicious Row  45.3
3 Red Delicious Row  42.1
```

- What are the levels in our factor (row)?

```
apple_harvest |> n_pct(row)
```

```
      row      n (pct)
Honeycrisp Row 18 (33.3%)
Red Delicious Row 18 (33.3%)
Zap Apple Row 18 (33.3%)
```

- What are the summary statistics by level of row?

```
apple_harvest |> group_by(row) |> mean_median(bushels)
```

```
# A tibble: 3 × 4
  row      variable mean_sd median_iqr
<chr>    <chr>    <chr>    <chr>
1 Honeycrisp Row  bushels  41.6 (8.0) 41.0 (6.9)
2 Red Delicious Row bushels  40.6 (3.8) 42.0 (5.1)
3 Zap Apple Row   bushels  37.5 (4.4) 38.5 (5.1)
```

- Constructing the ANOVA table,

```
apple_harvest |> anovas(outcome = bushels,
                        grouping = row)
```

One-way ANOVA

| Source    | df | SS        | MS      | F     | p-value |
|-----------|----|-----------|---------|-------|---------|
| row       | 2  | 160.9693  | 80.4846 | 2.468 | 0.095   |
| Residuals | 51 | 1663.4383 | 32.6164 |       |         |

Hypotheses (row):

$H_0$ : All group means are equal (no effect of row)

$H_1$ : At least one group mean differs (effect of row exists)

Test statistic:  $F(2, 51) = 2.468$

p-value: 0.095

Conclusion: Fail to reject the null hypothesis ( $p = 0.095 \geq \alpha = 0.050$ )

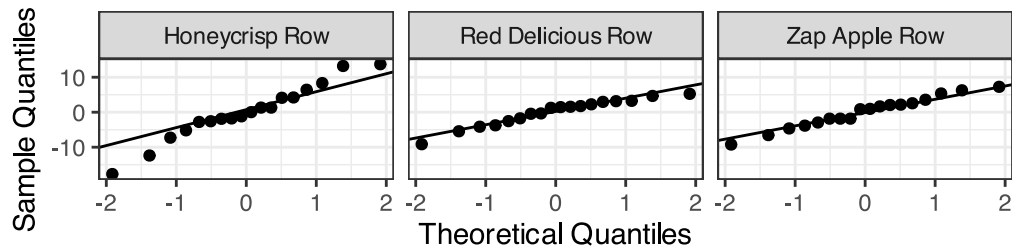
- **Hypotheses:**
- **Test Statistic and  $p$ -Value**
- **Rejection Region**
- **Conclusion and interpretation**

- Checking the normality of our Sweet Apple Acres dataset,

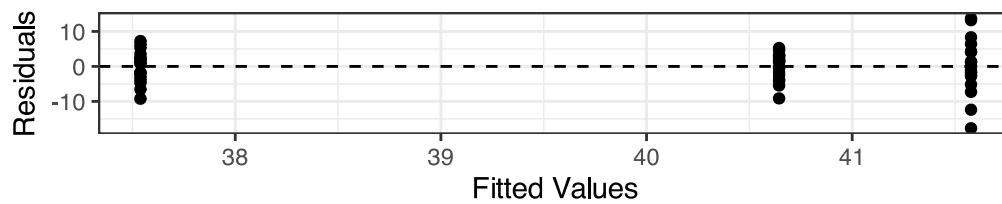
```
apple_harvest |> anova_check(outcome = bushels,
                             grouping = row)
```

## One-Way ANOVA Assumptions

### QQ Plots of Residuals by Group



### Residuals vs Fitted Values



- Applying the Kruskal-Wallis,

```
apple_harvest |> kruskal_wallis(outcome = bushels,  
                              grouping = row)
```

Kruskal-Wallis test

M[Honeycrisp Row] = 41 (n = 18)

M[Red Delicious Row] = 42 (n = 18)

M[Zap Apple Row] = 38.45 (n = 18)

Hypotheses:

$H_0$ : All group distributions are equal (no effect of row)

$H_1$ : At least one group distribution differs in location (effect of row exists)

Test statistic:  $H(2) = 5.953$

p-value: 0.051

Conclusion: Fail to reject the null hypothesis ( $p = 0.051 \geq \alpha = 0.050$ )

- **Hypotheses:**

- **Test Statistic and  $p$ -Value**

- **Rejection Region**

- **Conclusion and interpretation**

## Example 2: Rarity's Boutique

- Rarity flits between racks of fabric: “Darling, I need your professional opinion. Between my Canterlot Couture line, my everyday Ponyville Casual line, and my absolutely *dazzling* Gala Elegance line, I use wildly different amounts of gemstones. Is the difference really as dramatic as I think?”

```
# A tibble: 3 × 2
  collection      gem_count
  <chr>          <dbl>
1 Canterlot Couture    59
2 Canterlot Couture    57
3 Canterlot Couture    64
```

- We will use the `boutique_gems` dataset to answer the following question:
  - Does the number of gems used per dress differ across the three collections?
- What are the levels in our factor (`collection`)?

```
boutique_gems |> n_pct(collection)
```

```
      collection    n (pct)
Canterlot Couture 15 (33.3%)
  Gala Elegance 15 (33.3%)
Ponyville Casual 15 (33.3%)
```

- What are the summary statistics by level of collection?

```
boutique_gems |> group_by(collection) |> mean_median(gem_count)
```

```
# A tibble: 3 × 4
  collection      variable mean_sd    median_iqr
  <chr>          <chr>    <chr>    <chr>
1 Canterlot Couture gem_count 55.9 (8.4) 57.0 (8.0)
2 Gala Elegance    gem_count 124.3 (7.6) 124.0 (8.5)
3 Ponyville Casual  gem_count 87.9 (6.4) 87.0 (9.0)
```

- Constructing the ANOVA table,

```
boutique_gems |> anovas(outcome = gem_count,
                        grouping = collection)
```

One-way ANOVA

| Source     | df | SS         | MS         | F       | p-value |
|------------|----|------------|------------|---------|---------|
| collection | 2  | 35140.5778 | 17570.2889 | 311.110 | < 0.001 |
| Residuals  | 42 | 2372.0000  | 56.4762    |         |         |

Hypotheses (collection):

$H_0$ : All group means are equal (no effect of collection)

$H_1$ : At least one group mean differs (effect of collection exists)

Test statistic:  $F(2, 42) = 311.11$

p-value: < 0.001

Conclusion: Reject the null hypothesis ( $p = < 0.001 < \alpha = 0.050$ )

- **Hypotheses:**

- **Test Statistic and  $p$ -Value**

- **Rejection Region**

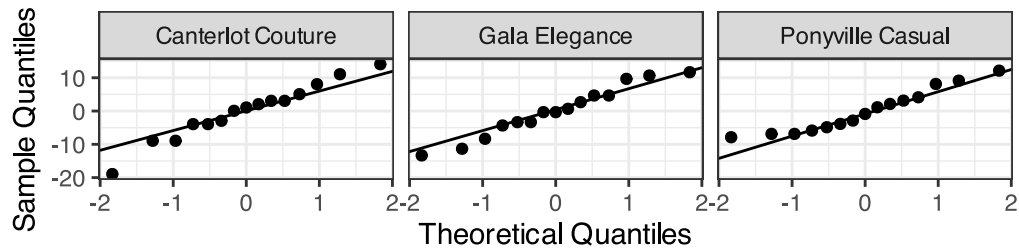
- **Conclusion and interpretation**

- Checking the normality of Rarity's dataset,

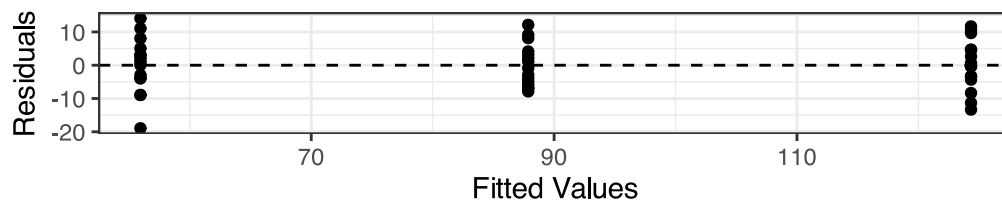
```
boutique_gems |> anova_check(outcome = gem_count,
                             grouping = collection)
```

## One-Way ANOVA Assumptions

### QQ Plots of Residuals by Group



### Residuals vs Fitted Values



- Applying the Kruskal-Wallis,

```
boutique_gems |> kruskal_wallis(outcome = gem_count,
                                grouping = collection)
```

Kruskal-Wallis test

M[Canterlot Couture] = 57 (n = 15)

M[Gala Elegance] = 124 (n = 15)

M[Ponyville Casual] = 87 (n = 15)

Hypotheses:

$H_0$ : All group distributions are equal (no effect of collection)

$H_1$ : At least one group distribution differs in location (effect of collection exists)

Test statistic:  $H(2) = 39.148$

p-value:  $< 0.001$

Conclusion: Reject the null hypothesis ( $p = < 0.001 < \alpha = 0.050$ )

- **Hypotheses:**

- **Test Statistic and  $p$ -Value**

- **Rejection Region**

- **Conclusion and interpretation**

### Example 3: Ponyville Diner

- Sunny Skies wipes down the counter after the lunch rush: “Feels like Caramel always walks away with better tips than me and Lyra put together. Before I start complainin’ to the manager, can you check if there’s actually a difference in how much we’re bringing in?”

```
# A tibble: 3 × 2
  server    tip_bits
  <chr>      <dbl>
1 Sunny Skies  4.08
2 Sunny Skies  6.12
3 Sunny Skies  0.12
```

- We will use the `ponyville_diner_tips` dataset to answer the following question:
  - Does tip amount differ across the three servers?
- What are the levels in our factor (`server`)?

```
ponyville_diner_tips |> n_pct(server)
```

```
      server    n (pct)
  Caramel  22 (33.3%)
  Lyra Heartstrings 22 (33.3%)
  Sunny Skies 22 (33.3%)
```

- What are the summary statistics by level of server?

```
ponyville_diner_tips |> group_by(server) |> mean_median(tip_bits)
```

```
# A tibble: 3 × 4
  server      variable mean_sd median_iqr
  <chr>      <chr>      <chr>      <chr>
1 Caramel    tip_bits 6.7 (4.4) 6.0 (5.9)
2 Lyra Heartstrings tip_bits 4.4 (4.7) 2.9 (2.9)
3 Sunny Skies tip_bits 7.4 (8.9) 4.3 (8.5)
```

- Constructing the ANOVA table,

```
ponyville_diner_tips |> anovas(outcome = tip_bits,
                              grouping = server)
```

#### One-way ANOVA

| Source    | df | SS        | MS      | F     | p-value |
|-----------|----|-----------|---------|-------|---------|
| server    | 2  | 109.2923  | 54.6461 | 1.356 | 0.265   |
| Residuals | 63 | 2539.3768 | 40.3076 |       |         |

#### Hypotheses (server):

$H_0$ : All group means are equal (no effect of server)

$H_1$ : At least one group mean differs (effect of server exists)

Test statistic:  $F(2, 63) = 1.356$

p-value: 0.265

Conclusion: Fail to reject the null hypothesis ( $p = 0.265 \geq \alpha = 0.050$ )

- **Hypotheses:**

- **Test Statistic and  $p$ -Value**

- **Rejection Region**

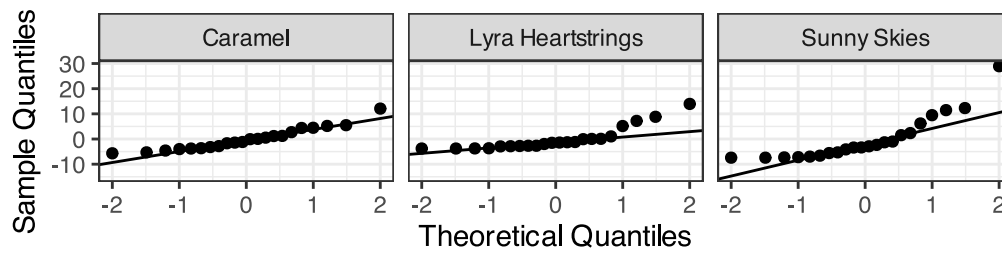
- **Conclusion and interpretation**

- Checking the normality of the Ponyville Diner dataset,

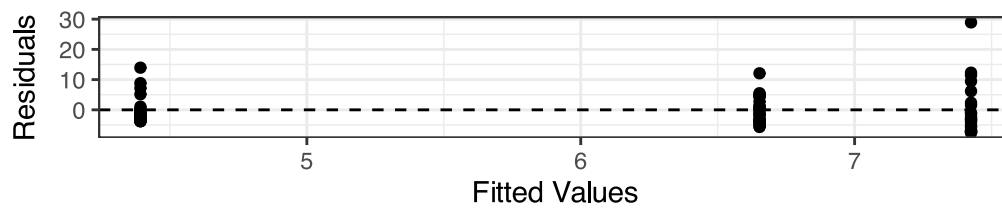
```
ponyville_diner_tips |> anova_check(outcome = tip_bits,  
                                   grouping = server)
```

## One-Way ANOVA Assumptions

### QQ Plots of Residuals by Group



### Residuals vs Fitted Values



- Applying the Kruskal-Wallis,

```
ponyville_diner_tips |> kruskal_wallis(outcome = tip_bits,  
                                     grouping = server)
```

Kruskal-Wallis test

M[Caramel] = 6.03 (n = 22)

M[Lyra Heartstrings] = 2.94 (n = 22)

M[Sunny Skies] = 4.305 (n = 22)

Hypotheses:

$H_0$ : All group distributions are equal (no effect of server)

$H_1$ : At least one group distribution differs in location (effect of server exists)

Test statistic:  $H(2) = 4.235$

p-value: 0.120

Conclusion: Fail to reject the null hypothesis ( $p = 0.120 \geq \alpha = 0.050$ )

- **Hypotheses:**

- **Test Statistic and  $p$ -Value**

- **Rejection Region**

- **Conclusion and interpretation**

## Let's Practice! 1: Wonderbolts Academy

- Fleetfoot drops a stack of stopwatch printouts on the table: “Command wants to know if lap times actually differ across our four training squads: Thunderhead, Stormwing, Skybolt, and Downburst.”

```
# A tibble: 3 × 2
  squad    lap_seconds
  <chr>      <dbl>
1 Thunderhead    60.4
2 Thunderhead    59.6
3 Thunderhead    62
```

- We will use the `wonderbolts_laptimes` dataset to answer the following question:
  - Do lap times differ across the four training squads?
- What are the levels in our factor (`squad`)?

```
wonderbolts_laptimes |> n_pct(squad)
```

```
   squad    n (pct)
Downburst 20 (25.0%)
Skybolt   20 (25.0%)
Stormwing 20 (25.0%)
Thunderhead 20 (25.0%)
```

- What are the summary statistics by level of `squad`?

```
wonderbolts_laptimes |> group_by(squad) |> mean_median(lap_seconds)
```

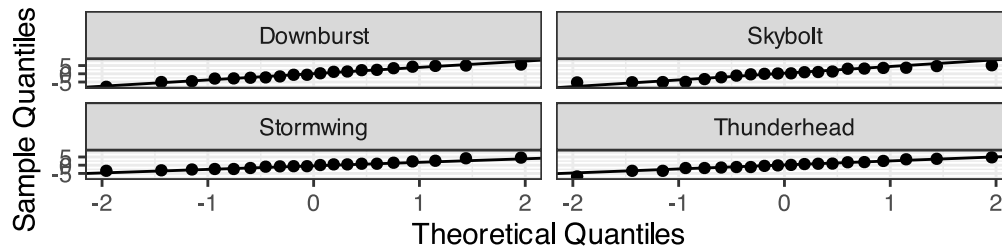
```
# A tibble: 4 × 4
  squad    variable  mean_sd  median_iqr
  <chr>      <chr>      <chr>      <chr>
1 Downburst lap_seconds 60.3 (3.7) 60.2 (5.2)
2 Skybolt   lap_seconds 53.1 (3.4) 53.3 (5.5)
3 Stormwing lap_seconds 61.2 (2.3) 61.0 (2.9)
4 Thunderhead lap_seconds 59.4 (2.8) 59.5 (3.3)
```

- Checking the normality,

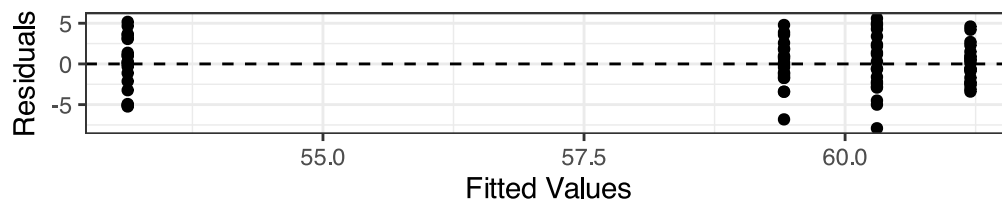
```
wonderbolts_laptimes |> anova_check(outcome = lap_seconds,
                                   grouping = squad)
```

## One-Way ANOVA Assumptions

### QQ Plots of Residuals by Group



### Residuals vs Fitted Values



- Constructing the ANOVA table,

```
wonderbolts_laptimes |> anovas(outcome = lap_seconds,
                               grouping = squad)
```

### One-way ANOVA

| Source    | df | SS       | MS       | F      | p-value |
|-----------|----|----------|----------|--------|---------|
| squad     | 3  | 804.4305 | 268.1435 | 28.098 | < 0.001 |
| Residuals | 76 | 725.2770 | 9.5431   |        |         |

### Hypotheses (squad):

$H_0$ : All group means are equal (no effect of squad)

$H_1$ : At least one group mean differs (effect of squad exists)

Test statistic:  $F(3, 76) = 28.098$

p-value: < 0.001

Conclusion: Reject the null hypothesis ( $p = < 0.001 < \alpha = 0.050$ )

- Applying the Kruskal-Wallis,

```
wonderbolts_laptimes |> kruskal_wallis(outcome = lap_seconds,
                                       grouping = squad)
```

Kruskal-Wallis test

M[Downburst] = 60.15 (n = 20)

M[Skybolt] = 53.35 (n = 20)

M[Stormwing] = 60.95 (n = 20)

M[Thunderhead] = 59.45 (n = 20)

Hypotheses:

$H_0$ : All group distributions are equal (no effect of squad)

$H_1$ : At least one group distribution differs in location (effect of squad exists)

Test statistic:  $H(3) = 38.039$

p-value:  $< 0.001$

Conclusion: Reject the null hypothesis ( $p = < 0.001 < \alpha = 0.050$ )

- After choosing the appropriate method,

- **Hypotheses:**

- **Test Statistic and  $p$ -Value**

- **Rejection Region**

- **Conclusion and interpretation**

## Let's Practice! 2: Klugetown Bazaar

- A weary trader waves you over between haggling matches: “Rustbolt, Glimmering, and Junker each swear their stall pulls in the best prices. I’ve got a season’s worth of sale records — do the three stalls actually sell for different amounts, or is everyone just talkin’ up their own wares?”

```
# A tibble: 3 × 2
  stall          sale_price_bits
  <chr>          <dbl>
1 Rustbolt's Relics          21.2
2 Rustbolt's Relics          18.9
3 Rustbolt's Relics          26.8
```

- We will use the `klugetown_bazaar` dataset to answer the following question:
  - Does sale price differ across the three market stalls?
- What are the levels in our factor (`stall`)?

```
klugetown_bazaar |> n_pct(stall)
```

```
      stall    n (pct)
Glimmering Wares 20 (33.3%)
Junker's Junction 20 (33.3%)
Rustbolt's Relics 20 (33.3%)
```

- What are the summary statistics by level of `stall`?

```
klugetown_bazaar |> group_by(stall) |> mean_median(sale_price_bits)
```

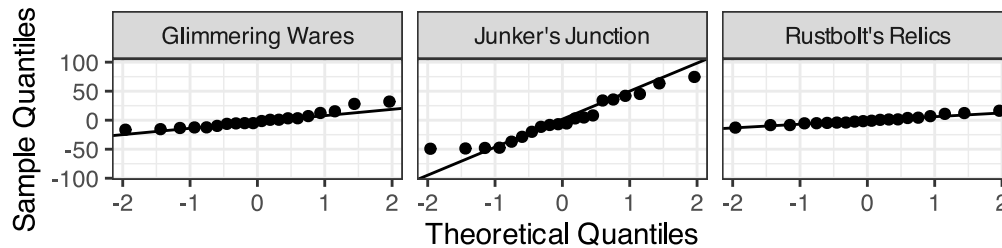
```
# A tibble: 3 × 4
  stall          variable    mean_sd    median_iqr
  <chr>          <chr>      <chr>      <chr>
1 Glimmering Wares sale_price_bits 39.6 (13.6) 36.4 (14.7)
2 Junker's Junction sale_price_bits 86.5 (38.5) 80.3 (65.1)
3 Rustbolt's Relics sale_price_bits 19.8 (7.4) 18.6 (8.7)
```

- Checking the normality,

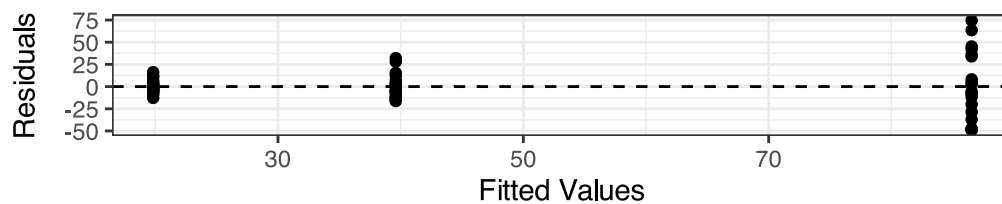
```
klugetown_bazaar |> anova_check(outcome = sale_price_bits,
                                grouping = stall)
```

## One-Way ANOVA Assumptions

### QQ Plots of Residuals by Group



### Residuals vs Fitted Values



- Constructing the ANOVA table,

```
klugetown_bazaar |> anovas(outcome = sale_price_bits,
                             grouping = stall)
```

### One-way ANOVA

| Source    | df | SS         | MS         | F      | p-value |
|-----------|----|------------|------------|--------|---------|
| stall     | 2  | 46965.2768 | 23482.6384 | 40.982 | < 0.001 |
| Residuals | 57 | 32661.2297 | 573.0040   |        |         |

### Hypotheses (stall):

$H_0$ : All group means are equal (no effect of stall)

$H_1$ : At least one group mean differs (effect of stall exists)

Test statistic:  $F(2, 57) = 40.982$

p-value: < 0.001

Conclusion: Reject the null hypothesis ( $p = < 0.001 < \alpha = 0.050$ )

- Applying the Kruskal-Wallis,

```
klugetown_bazaar |> kruskal_wallis(outcome = sale_price_bits,
                                   grouping = stall)
```

Kruskal-Wallis test

M[Glimmering Wares] = 36.36 (n = 20)  
 M[Junker's Junction] = 80.29 (n = 20)  
 M[Rustbolt's Relics] = 18.565 (n = 20)

Hypotheses:

$H_0$ : All group distributions are equal (no effect of stall)

$H_1$ : At least one group distribution differs in location (effect of stall exists)

Test statistic:  $H(2) = 43.348$

p-value:  $< 0.001$

Conclusion: Reject the null hypothesis ( $p = < 0.001 < \alpha = 0.050$ )

- After choosing the appropriate method,

- **Hypotheses:**

- **Test Statistic and  $p$ -Value**

- **Rejection Region**

- **Conclusion and interpretation**

### Let's Practice! 3: Baltimore Harbor

- The harbormaster spreads a stack of logbooks across the table: “Four dock crews unload every ship that comes through: Anchor, Tidewater, Saltmarsh, and Driftwood. Some of the sailors have been grumbling that one crew is dragging everyone’s schedule behind. Before I say anything to the crews, I want to know if unloading time actually differs across them.”

```
# A tibble: 3 × 2
  crew      unload_minutes
<chr>      <dbl>
1 Anchor Crew          1.3
2 Anchor Crew          2.19
3 Anchor Crew          5.13
```

- We will use the `baltimore_harbor` dataset to answer the following question:
  - Does unloading time differ across the four dock crews?
- What are the levels in our factor (`crew`)?

```
baltimore_harbor |> n_pct(crew)
```

```
      crew      n (pct)
Anchor Crew 20 (25.0%)
Driftwood Crew 20 (25.0%)
Saltmarsh Crew 20 (25.0%)
Tidewater Crew 20 (25.0%)
```

- What are the summary statistics by level of crew?

```
baltimore_harbor |> group_by(crew) |> mean_median(unload_minutes)
```

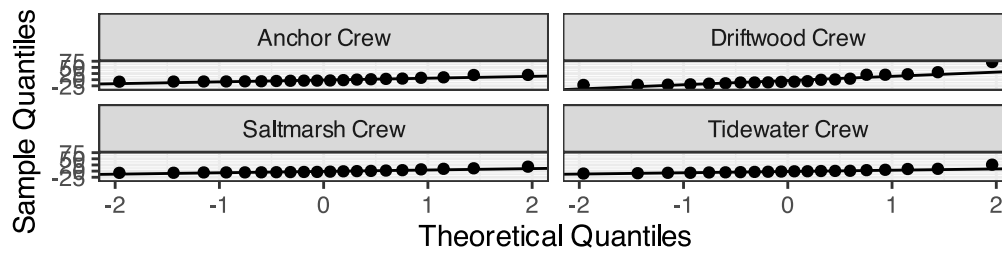
```
# A tibble: 4 × 4
  crew      variable      mean_sd      median_iqr
<chr>      <chr>      <chr>      <chr>
1 Anchor Crew  unload_minutes 8.9 (8.4)  5.4 (10.0)
2 Driftwood Crew  unload_minutes 22.1 (23.0) 13.7 (23.1)
3 Saltmarsh Crew  unload_minutes 7.8 (6.8)  5.1 (7.6)
4 Tidewater Crew  unload_minutes 10.8 (8.1)  9.5 (7.0)
```

- Checking the normality,

```
baltimore_harbor |> anova_check(outcome = unload_minutes,
                                grouping = crew)
```

## One-Way ANOVA Assumptions

### QQ Plots of Residuals by Group



### Residuals vs Fitted Values



- Constructing the ANOVA table,

```
baltimore_harbor |> anovas(outcome = unload_minutes,
                             grouping = crew)
```

### One-way ANOVA

| Source    | df | SS         | MS       | F     | p-value |
|-----------|----|------------|----------|-------|---------|
| crew      | 3  | 2616.1986  | 872.0662 | 4.906 | 0.004   |
| Residuals | 76 | 13509.4136 | 177.7554 |       |         |

### Hypotheses (crew):

$H_0$ : All group means are equal (no effect of crew)

$H_1$ : At least one group mean differs (effect of crew exists)

Test statistic:  $F(3, 76) = 4.906$

p-value: 0.004

Conclusion: Reject the null hypothesis ( $p = 0.004 < \alpha = 0.050$ )

- Applying the Kruskal-Wallis,

```
baltimore_harbor |> kruskal_wallis(outcome = unload_minutes,
                                   grouping = crew)
```

Kruskal-Wallis test

M[Anchor Crew] = 5.44 (n = 20)

M[Driftwood Crew] = 13.67 (n = 20)

M[Saltmarsh Crew] = 5.09 (n = 20)

M[Tidewater Crew] = 9.475 (n = 20)

Hypotheses:

$H_0$ : All group distributions are equal (no effect of crew)

$H_1$ : At least one group distribution differs in location (effect of crew exists)

Test statistic:  $H(3) = 7.917$

p-value: 0.048

Conclusion: Reject the null hypothesis ( $p = 0.048 < \alpha = 0.050$ )

- After choosing the appropriate method,

- **Hypotheses:**

- **Test Statistic and  $p$ -Value**

- **Rejection Region**

- **Conclusion and interpretation**